

Gap Verification Audit — $\ll(U_m\ll)$ Heaviside Amplifier and Job~B Scope (v5.78)

Star-Magic UQFF Audit

Gap Verification Audit — $\$U_m\$$ Heaviside Amplifier and Job B Scope

Paper: PAPER_1181

Session: 268

Status: Verification (read-only audit)

Supersedes: Repo-memory entry "UQFF_ug_equations_canonical" §"CRITICAL GAPS IDENTIFIED" item 1.

Abstract

A prior session-end summary listed two items as the next major work items in the Star-Magic UQFF codebase:

(i) the addition of a Heaviside phase-transition amplifier $\$(1+10^{13})\backslash,f_H)\cdot(1+f_q)\$$ to the universal-magnetism term $\$U_m\$,$ and (ii) an estimate that approximately 142 whitepapers required real authoring work under Job B (v5.78 closure propagation).

This audit verifies both claims directly against the source tree.

Claim (i) is **falsified**: the amplifier is already present in four independent locations, including the canonical `compute_Um_SOURCE4` of `MAIN_1_CoAnQi.cpp` (PAPER_421 integration).

Claim (ii) is **partially supported but mis-framed**: 113 papers require new authoring; 29 require verification only; together they sum to 142, but the work is not uniform.

1. Claim 1 — $\$U_m\$$ Heaviside Amplifier

1.1 The asserted canonical form

The asserted canonical form for the universal-magnetism term is

$$\begin{aligned} & \$\$ \\ & U_m \\ & = \Bigl[\sum_j \frac{\mu_j}{r_j} \Bigr] (1 - e^{-\gamma \cos(\pi t_n)}) \hat{\phi}_j \Bigr] \\ & \cdot P_{\mathrm{SCm}} \cdot E_{\mathrm{react}} \\ & \cdot \underbrace{(1+10^{13})\backslash,f_H)}_{\text{Heaviside}} \\ & \cdot \underbrace{(1+f_q)}_{\text{quasi-periodic}} , \\ & \$\$ \end{aligned}$$

where $\$f_H = \Theta(\rho_{\mathrm{SCm}} - \rho_c)\$$ activates the $\sim 10^{13}$ -fold amplification across the superconducting phase transition, and $\$f_q\$$ is a low-amplitude beating modulation tied to the 434-yr Gleisberg supercycle.

1.2 Direct source check

Location	File	Lines	Implements $(1+10^{13}f_H)\cdot(1+f_q)$?
SOURCE4 unified-field U_m	[MAIN_1_CoAnQi.cpp](../MAIN_1_CoAnQi.cpp#L24172-L24190)	24172–24190	**Yes** (PAPER\421)
SOURCE42 standalone PhysicsTerm			
[MAIN_1_CoAnQi.cpp](../MAIN_1_CoAnQi.cpp#L12100-L12170)	12100–12170	**Yes**	
compute_Um_Heaviside_factor()	[CondensedPhysics.py](../CondensedPhysics.py#L39950)	39950+	**Yes**
um_magnetic equation strings	[index.js](../index.js#L9759)	9759, 9899, 10018	**Yes**

The SOURCE4 implementation reads (verbatim, MAIN\1_CoAnQi.cpp L24180–24189):

```
```cpp
// PAPER_421: Heaviside phase-transition amplifier -- $(1 + 10^{13} \Theta(\rho_{SCm} - \rho_c))$
double f_H = (body.SCm_density >= rho_c_SOURCE4) ? 1.0 : 0.0;
double heaviside_factor = 1.0 + 1e13 f_H;

// PAPER_421: Quasi-periodic beating modifier -- $(1 + A_q \cos(\Delta\omega t))$
// $\Delta\omega = 2\pi/(434365.25)$ rad/day encodes the 434-year Gleisberg supercycle beat
double quasi_factor = 1.0 + A_q_SOURCE4 std::cos(Delta_omega_SOURCE4 t);

return Um_base heaviside_factor quasi_factor;
```
```

with constants declared at MAIN\1_CoAnQi.cpp L24016–L24018:

```
```cpp
const double rho_c_SOURCE4 = 1e15; // kg/m³
const double A_q_SOURCE4 = 0.1; // 10%
const double Delta_omega_SOURCE4 = 2.0 PI_SOURCE4 / (434.0365.25); // rad/day
```
```

1.3 Verdict

Claim 1 is **falsified**. The amplifier exists in:

- the SOURCE4 unified-field assembly (compute_Um_SOURCE4),
- the SOURCE42 catalog (HeavisideFactor, HeavisideUmWithHeaviside),
- the Python calculator stack (compute_Um_Heaviside_factor),
- and the JavaScript library's documented equation strings.

The repository-memory note flagging the gap is **stale**; it predates the PAPER\421 integration that added the amplifier to the SOURCE4 namespace.

1.4 Recommended follow-up (optional, low priority)

Although the amplifier is implemented, four refinements remain worth auditing:

1. Confirm QCalcGeom.py v2.1.0's F_U assembly threads the amplified U_m into the simultaneous-equation solver (default path currently uses a placeholder $U_m=0$ at the habitable-zone crossing; this is physically correct for $\rho_{SCm} < \rho_c$ but should be explicitly tested).

2. Cross-check that `index.js`'s `calculateUm()` function uses the amplifier or only documents it in the metadata string.
3. Verify the $\rho_c = 10^{15} \text{ kg/m}^3$ threshold against Fermilab Type-II SC magnetar-core data (cf. [100_MUGE Compression cycle 3_Superconductive Resonance.txt](../100_MUGE%20Compression%20cycle%203_Superconductive%20Resonance.txt), L1476).
4. Confirm the 434-yr Gleisberg-scale $\Delta\omega$ matches the long-term solar-cycle anchor cited in [grok_share_cfdcad2f5.txt](../grok_share_cfdcad2f5.txt) §Document 7.

These are validation tasks, not implementation gaps.

2. Claim 2 — "~142 papers need real authoring"

2.1 Authoritative bucket tally

Direct enumeration of [job_b_categorization.csv](../job_b_categorization.csv) (1,199 rows, May 22 2026):

| Count | Bucket | Class |
|-------------|--|-------------|
| 982 | NO UPDATE (specific system / framework application) | scoped out |
| 42 | Add closing section: v5.78 27-decade vacuum ledger (PAPER_1170) + $\xi=13/3$ | authoring |
| 34 | NO UPDATE unless ledger/ ξ enters; verify only | conditional |
| 29 | Verify v5.78 closed-Lagrangian/ledger cross-ref present (Sess 225 baseline) | verify |
| 28 | Update tables; add P6–P14, CP4 #254–#264 | authoring |
| 20 | v5.78 closure block inserted; .tex regenerated (arXiv pdf _{latex}) | DONE |
| 15 | Add v5.78 closed-Lagrangian cross-ref (G1–G8) + CP4 #254 | authoring |
| 12 | Forward-pointer to PAPER_1174 P1–P14 + PAPER_1177–1180 | authoring |
| 11 | Forward-pointer to PAPER_1171/1172 ($\xi=13/3$ R26+KK lock) | authoring |
| 10 | v5.78 closure block inserted; PDF regenerated | DONE |
| 5 | .tex / PDF regenerated + SOURCE REPAIR (Unicode) | DONE |
| 5 | Add three-anchor SI closure banner (Sess 237–241); STRUCTURAL | authoring |
| 5 | .tex converted from .md; PDF regenerated | DONE |
| 1 | duplicate row (stale) | drop |
| 1199 | TOTAL | — |

2.2 Reclassified work load

| Class | Count | Notes |
|-----------------------------------|-------------|--------------------------------------|
| Authoring required | 113 | 42 + 28 + 15 + 12 + 11 + 5 |
| Verification only | 29 | confirm existing cross-ref |
| Conditional (audit, no-op likely) | 34 | trigger only if ξ /ledger enters |
| Already DONE v5.78 | 35 | 20 + 10 + 5 |
| Out of scope (NO UPDATE) | 982 | scoped-out by topic |
| Stale duplicate | 1 | drop |
| Total | 1199 | |

2.3 Verdict

Claim 2 is **partially supported but imprecise**.

$\$113 + 29 = 142$ papers will be **touched** by Job B (authored or verified), but the prior framing " ~ 142 papers needing real authoring" overstates the authoring fraction by a factor of $\$142/113 \approx 1.26$. The honest split is **113 author + 29 verify**.

3. Revised Next-Step Priority Ranking

Given that the $\$U_m$ Heaviside gap is closed and the Job B count is 113 (not 142), the previously published 8-step plan should be re-ordered:

1. **Delete** the $\$U_m$ amplifier step (was step 1) — already done.
2. **Generate** LEDGER_REVIEW_QUEUE.csv for the 43 flagged master_closures.csv rows (10 high-error $\$ge 10\%$ + 15 PARSE_FAIL + 18 NaN). *Was step 2; now step 1.*
3. **Recalibrate** the 10 high-residual ledger entries (IDs 270, 293, 330, 337, 351, 352, 739, 760, 764, 805) using v5.78 anchors (PAPER_1156, 1170, 1171, 1172). *Was step 3; now step 2.*
4. **Author** the 4–6 reusable v5.78 section templates (T- $\$Lambda$, T-LAG, T-SI, T-PRED, T- $\$xi$). *Was step 4; now step 3.*
5. **Bucket-batched paper updates** — 113 papers, 5–10 per commit, starting with the 42 cosmology / $\$Lambda$ / vacuum-energy papers using the T- $\$Lambda$ template. *Was step 5; now step 4.*
6. **Verify-only sweep** of the 29 bucket-G papers. *Was step 6; now step 5.*
7. **Regenerate** all updated PDFs with pdf_{latex}; commit .md + .pdf together.
8. **Final coherence audit** mirroring Session 261 (f6ba35a6).

4. Citations

- MAIN_1_CoAnQi.cpp L24016–24018, L24172–24190 (PAPER_421: Heaviside + quasi-periodic amplifiers in SOURCE4 $\$U_m$).
- MAIN_1_CoAnQi.cpp L12100–12170 (SOURCE42: HeavisideFraction, HeavisideFactor, HeavisideUmWithHeaviside).
- CondensedPhysics.py L39950+ (compute_Um_Heaviside_factor).
- index.js L9690, 9759, 9811, 9899, 9943, 10018 (f_heaviside, f_quasi, full um_magnetic equation strings).
- _job_b_categorization.csv (1,199 rows, May 22 2026).
- Job B brief: "Job B_Update papers with current canonical UQFF v5_78.txt" (395 lines, Phase B1–B4).
- COMPLETE_UQFF_EQUATIONS_REFERENCE.md L522–553 (canonical $\$U_m$ form).

5. Conclusion

The two items proposed as the "next big step" do not survive direct source-tree verification in the form they were stated:

- The $\$U_m$ Heaviside phase-transition amplifier is **already implemented**, in four places, with the canonical 10^{13} scale and the 434-yr Gleisberg-locked quasi-periodic beat; it is **not** the most-impactful uncoded physics.
- The Job B authoring queue is **113 papers**, not 142;

the 142 figure conflates authoring with verification.

The corrected next major step is the **ledger review queue** (43 flagged closures, with 10 high-residual recalibrations against the v5.78 anchors), followed by the 113-paper Job B authoring sweep using reusable templates.